

4-bit Up-down counter

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1 Objective

- The objective is to design, implement, and synthesize a 4-bit Up-Down counter using Verilog.
- The design is implemented on the RealDigital Boolean Board featuring a Xilinx Spartan-7 FPGA (XC7S50CSGA324-1).
- The counter increments or decrements at a rate of 1 Hz based on a switch-controlled mode and can be asynchronously reset via a push button.

2 Board

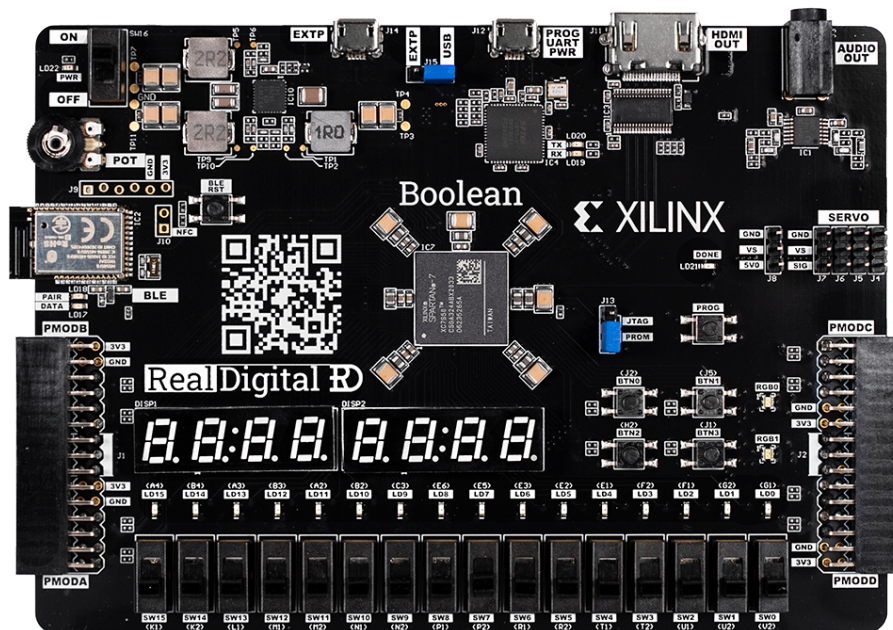


Diagram of board used

3 Verilog code

- To ensure observable counting, the 100 MHz system clock is stepped down to a 1 Hz enable signal.
- The module UpDownCounter consists of two main blocks:
 - A clock divider that counts up to 100,000,000 clock cycles to generate a 1 Hz pulse.
 - The counter logic that updates the LED states on that pulse.

```
1 'timescale 1ns / 1ps
2
3 module UpDownCounter (
4     input wire clk_100MHz,    // Main 100 MHz system clock
5     input wire reset,         // Reset button
6     input wire mode,          // Control switch: 1 = UP, 0 = DOWN
7     output reg [3:0] sCount   // 4-bit output to LEDs
8 );
9 // 1. Clock Divider (1 Hz Enable Generator)
10 reg [26:0] counter_1Hz = 0;
11 wire enable_1Hz;
12
13 always @(posedge clk_100MHz or posedge reset) begin
14     if (reset) begin
15         counter_1Hz <= 0;
16     end else if (counter_1Hz == 100000000 - 1) begin
17         counter_1Hz <= 0;
18     end else begin
19         counter_1Hz <= counter_1Hz + 1;
20     end
21 end
22
23 assign enable_1Hz = (counter_1Hz == 100000000 - 1) ? 1'b1 : 1'b0;
24
25 // 2. 4-bit Up/Down Counter Logic
26 always @(posedge clk_100MHz or posedge reset) begin
27     if (reset) begin
28         sCount <= 4'b0000;
29     end else if (enable_1Hz) begin
30         if (mode == 1'b1) begin
31             sCount <= sCount + 1'b1; // Count UP
32         end else begin
33             sCount <= sCount - 1'b1; // Count DOWN
34         end
35     end
36 end
37
38 endmodule
```

Listing 1: Verilog Source Code

4 Hardware Mapping

The hardware components are mapped to the FPGA physical pins as detailed in the below Table.

Signal Name	Component Type	Board Label	FPGA Pin
clk_100MHz	System Clock	Internal	F14
mode	Slide Switch	SW0	V2
reset	Push Button	BTN0	J2
sCount[3]	LED (MSB)	LD3	F2
sCount[2]	LED	LD2	F1
sCount[1]	LED	LD1	G2
sCount[0]	LED (LSB)	LD0	G1

Pin assignments for the RealDigital Boolean Board

5 Constraints File (XDC)

The Xilinx Design Constraints (.xdc) file bridges the logical ports defined in the Verilog code to the physical architecture of the Boolean board.

```
1 # === 100 MHz System Clock ===
2 set_property -dict { PACKAGE_PIN F14 IOSTANDARD LVCMOS33 } [get_ports {
   clk_100MHz }];
3 create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [
   get_ports { clk_100MHz }];
4 # Maps 'mode' to Switch SW0
5 set_property -dict { PACKAGE_PIN V2 IOSTANDARD LVCMOS33 } [get_ports {
   mode }];
6
7 # Maps 'reset' to Button BTN0
8 set_property -dict { PACKAGE_PIN J2 IOSTANDARD LVCMOS33 } [get_ports {
   reset }];
9
10 # Maps sCount to LEDs: LD3, LD2, LD1, LD0
11 set_property -dict { PACKAGE_PIN F2 IOSTANDARD LVCMOS33 } [get_ports {
   sCount[3] }];
12 set_property -dict { PACKAGE_PIN F1 IOSTANDARD LVCMOS33 } [get_ports {
   sCount[2] }];
13 set_property -dict { PACKAGE_PIN G2 IOSTANDARD LVCMOS33 } [get_ports {
   sCount[1] }];
14 set_property -dict { PACKAGE_PIN G1 IOSTANDARD LVCMOS33 } [get_ports {
   sCount[0] }];
```

Listing 2: Boolean Board Constraints (boolean.xdc)

6 Functions of input and output observed

- **Reset (BTN0):** This push button is used to reset the counter asynchronously. When pressed, the counter is immediately set to 0000_2 , irrespective of the clock or mode.
- **Mode (SW0):** This slide switch controls the direction of counting:
 - Mode = 1: Counter increments (Up-counting)
 - Mode = 0: Counter decrements (Down-counting)

6.1 Observed Output

- For Mode = 1, the LED sequence follows:

$0000 \rightarrow 0001 \rightarrow 0010 \rightarrow \dots \rightarrow 1111 \rightarrow 0000$

- For Mode = 0, the LED sequence follows:

$0000 \rightarrow 1111 \rightarrow 1110 \rightarrow \dots \rightarrow 0001 \rightarrow 0000$

- The output is recorded and uploaded in the link provided below

6.2 Output Link

Please click the following link for the output
4-Bit-Up-Down Counter